

# Extended Abstract

**Motivation** Protein-DNA interactions are one of the most prominent interactions in biology: they allow cells to "read" DNA, by using DNA-binding protein called transcription factors (TFs), that bind specific stretches of DNA for activation and repression of nearby genes. Yet, despite its significance, our understanding of protein-DNA interactions is limited. A number of experimental methods can capture the DNA-binding interactions of specific proteins, however these experiments are expensive and data can be sparse for poorly studied TFs / mutant variants. Overall, there is still no known direct mapping from protein sequence to DNA sequence, and vice versa.

**Method** We present three novel contributions to the study of generating DNA sequences to bind to protein targets. First, we curate a high quality dataset of Protein Binding Microarray (PBM) data from over 3,500 experiments, each containing roughly 40,000 DNA sequences and one protein binding domain. Second, we present a novel architecture that leverages two pretrained biological foundation models (a DNA language model and a protein language model) to preform protein-conditioned DNA sequence generation by fusing modality-specific embeddings with a cross-attention layer. Third, we frame the fluorescence intensities from a PBM experiment as a preference optimization problem and show that Weighted DPO is a biophysically justified method for modeling the preference between two protein-DNA binding states.

**Implementation** Our work demonstrates significant engineering implementation. We built a pipeline to train the model in two phases: a warmstart supervised finetuning (SFT) phase, and a weighted DPO finetuning phase. The architecture, training loop, and data manipulation was written from scratch using PyTorch Lightning and automatically handles both phases of train, validation, test, and inference. Additionally, we implemented a number of optimizations, including the pre-computation of reference likelihoods used in DPO to eliminate the memory load of maintaining a reference model.

**Results** We compare the performance of two models: one finetuned on top of Evo 2 and one using a *ab initio* transformer. We find that training on top of Evo 2 leads to instability, while an *ab initio* transformer model achieves lower training loss. We perform further validation by manually inspecting the model's generations when conditioned on a number of well-studied human and mouse transcription factors and find that the model outputs contain the canonical DNA motifs that the TFs are known to bind to, proving the effectiveness of our warm start SFT phase. We also demonstrate that DPO post-training is able to further improve the accuracy of our model on assigning higher likelihoods of generation to held-out positive binding DNA sequences compared to held-out negative binding DNA sequences.

**Discussion** Our results we show that deep learning is an effective and promising method to better understand protein-DNA binding interactions. However, our method is still bottlenecked by compute, utilizing less than 1% of the total PBM data that exists in our dataset. Furthermore, we encountered issues with finetuning on top of Evo 2, showcasing how large foundation models can yield embeddings that are difficult to work with and can cause exploding gradient norms. Finally, we highlight how current biological foundation model infrastructure can be difficult to work with.

**Conclusion** Despite these limitations, our work is a promising first step towards generating custom DNA sequences to recruit and bind to specific protein targets. As the performance of our method continues to improve, it could one day open the door for improved gene therapy treatments, allowing biologists to design targeted DNA sequences that can stimulate or repress the activity of specific genes in specific cell types. Such an ability would improve drug efficacy, limit off-target side effects, and boost the specificity of effect.

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# Capturing Protein-DNA interactions via Weighted Direct Preference Optimization

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## Abstract

Protein-DNA interactions are one of the most prominent interactions in biology, allowing cells to "read" DNA. We present three novel contributions to the study of generating DNA sequences to bind to protein targets. First, we curate a high quality dataset of Protein Binding Microarray (PBM) data from over 3,500 experiments, each containing roughly 40,000 DNA sequences and one protein binding domain. Second, we present a novel architecture that leverages two pretrained biological foundation models (a DNA language model and a protein language model) to preform protein-conditioned DNA sequence generation by fusing modality-specific embeddings with a cross-attention layer. Third, we frame the fluorescence intensities from a PBM experiment as a preference optimization problem and show that Weighted DPO is a biophysically justified method for modeling the preference between two protein-DNA binding states. Through our training, we show that deep learning methods can learn biologically meaningful protein and DNA sequences, and improve accuracy on prioritizing positive DNA binding sequences for a given protein.

## 1 Introduction

Protein-DNA interactions are one of the most prominent interactions in biology—fundamentally, it is how cells “read” DNA. Given a mutation in DNA (e.g., an ‘A’ to a ‘C’), protein-DNA interactions connect the change into a unique trait (e.g., height, hair color, lactose intolerance, or even sometimes, cancer). Specifically, a family of proteins called transcription factors (TFs)—one of the largest human protein families, making up ~10% of all human proteins—bind to specific stretches of DNA. [Vaquerizas et al. (2009)] (e.g., protein ‘JUN-B’ preferentially binds to DNA sequence ‘TGACTCA’ [Colin et al. (2011)]). This allows the TF protein to locate a specific part of the genome, turn on or off transcription of nearby genes into mRNA, that is in turn translated into protein, and goes on to perform its designated biological function. This is often referred to as the “Central Dogma of Biology”, shared across all kingdoms of life. [Crick (1970)]

Given the significance of protein-DNA interactions, many experimental methods have been developed to capture the DNA binding interactions of specific proteins. To capture isolated protein-DNA interactions in an *in vitro* setting, SELEX (Systematic Evolution of Ligands by Exponential Enrichment) and PBM (Protein binding microarray) and its variants are popular assays [Gold (2015), Berger and Bullyk (2009)], while for cellular *in vivo* settings, ChIP-Seq (Chromatin ImmunoPrecipitation Sequencing)

and its variants are the gold standard [Park (2009)]. Traditional methods have then attempted to collect and record the most representative DNA binding sequence of well-known TFs from these experiments as hand-curated public databases [Rauluseviciute et al. (2024), Vorontsov et al. (2024)]. However, for poorly studied TFs or mutant variants (e.g., TF with a single amino acid mutation), this approach fails. And despite the significance of protein-DNA interactions, there is still no known direct mapping of protein sequence-to-DNA sequence, or vice versa.

Recently, taking inspiration from natural language, there have been successes modeling both proteins and DNA molecules via deep learning language models (LM). Interestingly, proteins and nucleic acids are sequential biomolecules that are uniquely and sufficiently encoded as 1-dimensional strings: in the case of proteins, as a string of amino acids (e.g., RIKVERKRLRNRLAATKCRKRKLERI-ARLEDKVKTLKAENAGLSSTAGLLREQVAQLKQKVMTH codes for the DNA-binding domain of TF JUN-B; where each letter codes for 1 of 20 amino acids), and for nucleic acids, as a string of nucleotides (e.g., TGACTCA codes for the DNA that JUN-B binds to; where each letter codes for 1 of 4 nucleotides). This allows proteins and DNA to be modeled in a language-like fashion, trained on self-supervised hidden token prediction tasks. To model proteins, protein language models (PLMs) have been trained to predict hidden amino acid tokens across all known natural proteins (~3 billion proteins), and have shown to capture natural patterns of protein evolution [Hayes et al. (2025)]. Similarly, to model DNA—the single, foundational source of information for all living organisms—DNA language models (DNA-LMs) have been trained to predict hidden nucleotide tokens on genomes across all domains of life (~9 trillion sequences), and have begun to capture some key underlying genetic drivers [Brix et al. (2025)].

Here, extending the successes of PLMs and DNA-LMs, we aim to model protein-DNA interactions: by aligning a pretrained PLM and a pretrained DNA-LM via a weighted, direct preference optimization (DPO) objective from experimental PBM data, to generate binding DNA sequences, conditioned on a DNA binding protein sequence.

## 2 Related Work

### 2.1 Database

In order to map proteins to DNA, traditionally, the binding sequence affinity of well-known proteins have been collected from various experiments and curated into publicly available databases. JASPAR is one of the largest and most widely used publicly accessible database of manually curated DNA binding sequences of eukaryotic transcription factors, containing binding sequence profiles for over 5,000 proteins [Rauluseviciute et al. (2024)]. HOCOMOCO is another widely used publicly accessible database, focused on human and mouse transcription factors, containing binding sequence profiles for around 1,600 human and mouse proteins [Vorontsov et al. (2024)]. However, for all databases, the database directly maps one protein sequence to one (or a few) DNA sequence(s); therefore, for example, when introduced with an unknown protein, or even a known transcription factor with a single amino acid mutation, its binding DNA sequence is quickly unavailable, without any generalizable mapping.

### 2.2 Structure

More recently, given successes in protein structure models, there have been methods to directly predict the structure of a DNA-bound protein complex, which is in turn used to predict the validity and strength of interaction. The latest of the AlphaFold suite of models, AlphaFold3 [Abramson et al. (2024)], can predict the structure of protein-ligand complexes (where here, the ligand is DNA), while the latest RoseTTAFold2 suite of models includes a designated protein-DNA structure model, RoseTTAFold2NA [Baek et al. (2024)]. However, previous studies have shown that structure models are mostly trained on positive examples and can struggle to predict negative binding complexes. Additionally, even with a good structure prediction, predicting binding affinity is a nontrivial task.

### 2.3 Biological Language Models

Most recently, advances in natural language models have extended to modeling sequential biomolecules, including proteins and DNAs:

### 2.3.1 Protein Language Models (PLMs)

PLMs have shown to successfully capture the biological function of proteins from sequence. Trained across all known natural proteins, the suite of ESM models have shown to capture the patterns and preferences of protein evolution over time [Lin et al. (2023)], and the underlying embeddings have been shown to capture structural information, that can be used to predict protein structure, as well as functional information, that can be used to design proteins with desired functions [Meier et al. (2021); Im et al. (2025); Hsu et al. (2022)]

### 2.3.2 DNA Language Models (DNA-LMs)

DNA-LMs have been trained on genomes across all domains of life. The suite of Evo models have shown to capture evolutionary patterns of DNA, across both prokaryotes and eukaryotes, by training on an extensive range of genomes across the tree of life [Nguyen et al. (2024)]. The underlying embeddings have been shown to capture different types of functional DNA (e.g., protein coding, regulatory, splicing), as well as design of new DNA systems from molecular scale (e.g., CRISPR-Cas) to genomic scale (e.g., human mitochondrial, yeast) [Brix et al. (2025)].

## 3 Method

### 3.1 Weighted DPO

Direct Preference Optimization (DPO) is an implicit reinforcement learning (RL) method commonly used to align a language model on human-labeled pairs of preference data [Rafailov et al. (2024)]. It is widely adopted in natural language to finetune a base pretrained language models towards specific human preference, and has supplemented (and often superseded) previous finetuning methods, such as the previous generation of Proximal Policy Optimization (PPO) / Reinforcement Learning with Human Feedback (RLHF) methods [Ouyang et al. (2022)].

Prior work has successfully extended DPO to protein language models: The latest of the ESM suite of models, ESM3 was finetuned on a DPO-like objective (IRPO; Iterative Reasoning Preference Optimization) on predicted structure confidence [Hayes et al. (2025)], while in ProteinDPO, the ESM inverse folding model, ESM-IF1 was finetuned with a weighted DPO objective on experimental thermal stability data. [Widatalla et al. (2024)]

Here, we apply weighted DPO on our model: an autoregressive DNA-LM, conditioned on a protein sequence. Surprisingly, we find that the weighted DPO objective captures the true free energy difference between two protein-DNA bound states, that directly follow the biophysical thermodynamics of the binding reaction. To show this, we review the original DPO method [Ouyang et al. (2022)]:

**Deriving the DPO objective** We start with the standard RL objective for language models from RLHF, where for prompt  $x$  and completion  $y$ , the objective is to maximize the general reward function  $r$ , while minimizing deviation away from the original reference model via a Kullback-Leibler Divergence (KL divergence) regularization:

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi} [r(x, y)] - \beta D_{\text{KL}} [\pi(y | x) \parallel \pi_{\text{ref}}(y | x)] \quad (1)$$

From DPO, we are able to show that there exists a closed-form solution to the optimization objective, described by the optimal policy  $\pi^*$ , and its corresponding optimal reward function  $r^*$ , and partition function  $Z$ :

$$\pi^*(y | x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y | x) \exp \left( \frac{1}{\beta} r(x, y) \right), \quad (2)$$

$$r^*(x, y) = \beta \log \frac{\pi_r^*(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x) \quad (3)$$

**Natural language preference** Next, in the original DPO work, human preference between two completions,  $y_1, y_2$  is assumed to follow a Bradley-Terry model: a probabilistic preference for one

completion over another, given the same prompt  $p(y_1 > y_2 | x)$ , described by the difference in some latent reward between the two completions  $r_\phi(y_1 | x) - r_\phi(y_2 | x)$ :

$$p(y_1 > y_2 | x) = \sigma(r_\phi(y_1 | x) - r_\phi(y_2 | x)) = \frac{\exp(r_\phi(y_1 | x))}{\exp(r_\phi(y_1 | x)) + \exp(r_\phi(y_2 | x))} \quad (4)$$

Furthermore, the assumption is that the true label for human preference is hard (i.e., there is an absolute preferred state, of 1 or 0):

$$y_1 > y_2 = \mathbf{1}_{\{y_1 > y_2\}} \quad (5)$$

Therefore, optimizing to solve the preference classification task with hard, one-hot binary labels, the resulting binary cross entropy loss results in the original DPO loss:

$$\begin{aligned} \mathcal{L}_{\text{BCE}} &= -\mathbb{E}_p[\log \hat{p}] \\ &= -\mathbb{E}_{(x, y_1, y_2) \sim \mathcal{D}} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \beta \log \frac{\pi_\theta(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right) \right] \\ &= \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) \end{aligned} \quad (6)$$

**Protein-DNA interaction preference** However, in the case of protein-DNA binding, when applying the same assumptions, we find that the DPO objective captures some true biophysical properties.

In our model: an autoregressive DNA-LM  $\pi$  generates a pair of DNA sequences  $y_1, y_2$ , conditioned on a protein sequence prompt  $x$ . We want to optimize for DNA sequences that the protein  $x$  binds more strongly to  $y_1 > y_2 | x$ .

Recall: Given a system with  $i \in M$  possible states, each with free energy  $G_i$ , the probability that the system will be in state  $i$  is described by the Boltzmann distribution (assuming a canonical ensemble; constant  $N, V, T$ ) [Boltzmann (1868)]:

$$p_i = \frac{1}{Z} \exp \left( -\frac{G_i}{kT} \right) = \frac{\exp \left( -\frac{G_i}{kT} \right)}{\sum_{j=1}^M \exp \left( -\frac{G_j}{kT} \right)} \quad (7)$$

Assuming a simple, two-state system, where State 1: Protein bound to DNA<sub>1</sub>, State 2: Protein bound to DNA<sub>2</sub>, each with free energy  $\Delta G_1, \Delta G_2$ , we can describe the probability that the system will be in State 1 over State 2:

$$\text{Protein-DNA}_1 (+ \text{DNA}_2) \xrightleftharpoons[\Delta G_1]{\Delta G_2} \text{Protein-DNA}_2 (+ \text{DNA}_1) \quad (8)$$

$$p(\text{Protein-DNA}_1 > \text{Protein-DNA}_2) = \frac{\exp \left( -\frac{\Delta G_1}{kT} \right)}{\sum_{j=1}^M \exp \left( -\frac{\Delta G_j}{kT} \right)} \quad (9)$$

$$= \frac{\exp \left( -\frac{\Delta G_1}{kT} \right)}{\exp \left( -\frac{\Delta G_1}{kT} \right) + \exp \left( -\frac{\Delta G_2}{kT} \right)} \quad (10)$$

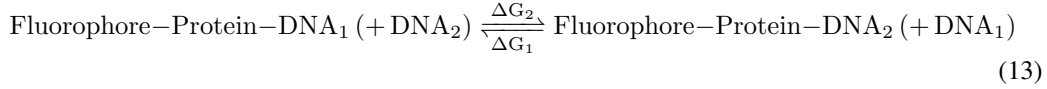
Note, that this is the same relationship as the Bradley-Terry model in Eq. 4. In other words, when applying the DPO loss in the case of protein-DNA interactions, the reward function directly captures the negative reduced energy  $(-\frac{\Delta G}{kT})$  of each paired state.

$$r(\text{Protein-DNA}_1) \approx \exp \left( -\frac{\Delta G_1}{kT} \right) \quad (11)$$

**Soft labels** Additionally, unlike the assumption in the original DPO, the true labels of the reaction preference are not hard (0 or 1). Instead, they are soft probabilities  $[0, 1]$ , that can be directly measured by experimental assays. With soft probability labels, the same binary cross entropy loss becomes the following:

$$\begin{aligned}\mathcal{L}_{BCE} &= -\mathbb{E}_p \left[ p_1 \log \sigma \left( \beta \log \frac{p_\theta(\mathbf{y}_1 | \mathbf{x})}{p_{\text{ref}}(\mathbf{y}_1 | \mathbf{x})} - \beta \log \frac{p_\theta(\mathbf{y}_2 | \mathbf{x})}{p_{\text{ref}}(\mathbf{y}_2 | \mathbf{x})} \right) \right. \\ &\quad \left. + (1 - p_1) \log \sigma \left( \beta \log \frac{p_\theta(\mathbf{y}_2 | \mathbf{x})}{p_{\text{ref}}(\mathbf{y}_2 | \mathbf{x})} - \beta \log \frac{p_\theta(\mathbf{y}_1 | \mathbf{x})}{p_{\text{ref}}(\mathbf{y}_1 | \mathbf{x})} \right) \right] \\ &= \mathcal{L}_{\text{Weighted DPO}}\end{aligned}\tag{12}$$

**Experimental PBM** In order to measure the soft probability labels between a pair of protein-DNA interactions, we use PBM experimental data. During PBM experiments, the protein is bound by a fluorophore, and the relative fluorescence is measured for each protein-DNA pair. In other words, we can view the reaction as follows:



Assuming the fluorophore does not affect the binding interaction between the protein and DNA, and assuming that the Beer-Lambert law holds when measuring fluorescence absorbance  $A_{\text{Fluorescence}}$  [Bouguer (1729)]:

$$A_{\text{Fluorescence}} \propto C_{\text{Fluorophore}} (= \varepsilon * b * C_{\text{Fluorophore}})\tag{14}$$

Given fluorescence absorbance  $A_{\text{Fluorescence}}$ , we can now measure relative concentration, and therefore, the relative probability of each protein-DNA pair:

$$p(\text{Protein-DNA}_1 > \text{Protein-DNA}_2) \approx \frac{C_{\text{Protein-DNA}_1}}{C_{\text{Protein-DNA}_1} + C_{\text{Protein-DNA}_2}} = \frac{A_{\text{Protein-DNA}_1}}{A_{\text{Protein-DNA}_1} + A_{\text{Protein-DNA}_2}}\tag{15}$$

### 3.2 Model Architecture

In order to design an DNA-LM conditioned on protein sequence, we constructed the following model architecture:

#### 3.2.1 DNA Encoder

We used two DNA encoders:

1. Evo2-7b: The latest of the Evo suite of DNA-LMs (7 billion parameters), pretrained with 8192 context length on genomes across all of life. The model uses a long convolution and attention-based StripedHyena2 architecture. Unfortunately, the current publicly available version of Evo 2 doesn't allow for finetuning, so we instead add an attentional causal transformer layer on top of Evo 2 to extend the model and finetune the embeddings.
2. *ab initio* Attention transformer: As described below, we also try to train our own transformer DNA encoder from scratch. We use a 3 layer transformer with Rotary Positional Embeddings, GELU activation, LayerNorm, and a ParallelGatedMLP output head to match the architecture of the transformer layers in Evo 2.

#### 3.2.2 Protein Encoder

We selected ESM-Cambria (ESM-C), the latest in the suite of ESM models and currently considered state of the art, for our pretrained PLM. ESM-C is a 6 billion parameter model trained on masked

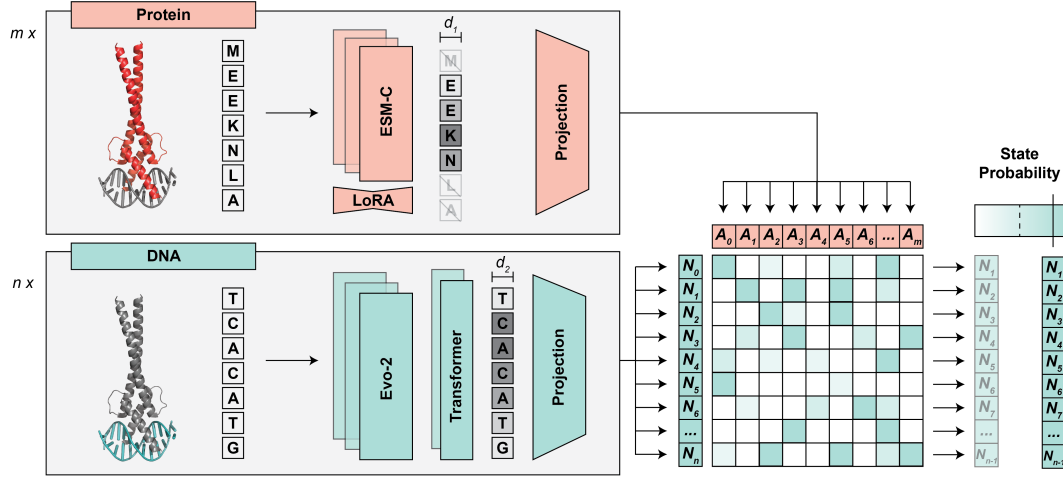


Figure 1: Model Architecture

token prediction across over 3.2 billion proteins. We finetune ESM by applying low rank adaptation (LoRA) parameters to the query linear layers in the last six attention layers.

### 3.2.3 Cross Attention

In order to condition the generation of DNA sequences on protein sequence, we placed a cross-attention layer after the modality-specific encoders. The cross-attention layer takes as query the DNA embedding, and applies key and value conditioning from the protein embedding. The output is a DNA embedding, conditioned on a protein embedding.

### 3.2.4 Outputs

In order to do causal next token prediction, we take the protein-conditioned DNA embedding, pass it through a language modeling head, and obtain DNA logits. Additionally, we include a regularization protein task to keep the protein embeddings intact. We directly take the protein embeddings from ESM-C, pass through a separate language model projection head to obtain protein logits, to predict masked tokens.

## 3.3 Data

For training data, we collected a set of over 3,500 PBM experiments data from a publicly available database, CIS-BP (Catalog of Inferred Sequence Binding Preferences). Each experiment contains binding affinity for approximately ~40,000 DNA sequences, resulting in over 140 million unique protein-DNA binding interactions.

### 3.3.1 Source

In order to identify the free energy differences between binding states, we gathered data from experimental Protein Binding Microarrays (PBM) to create our preference data. PBM is a high-throughput, *in vitro* biological assay that can measure the binding affinity of many pairs of protein and DNAs in parallel.

At a high-level, the assay works as follows: (1) Each microarray device contains ~10,000s of microwells, with each surface printed with a unique library of DNA sequences. (2) A protein of interest, tagged with a fluorophore molecule, is expressed and purified, and prepared as a bulk solution. (3) The protein solution is passed over the microarray surface, allowing the protein to freely bind to DNA molecules bound to the surface. (4) The protein solution is washed away. (5) The remaining bound protein is detected via a fluorescence spectrometer. (6) The fluorescence intensity per well is recorded.

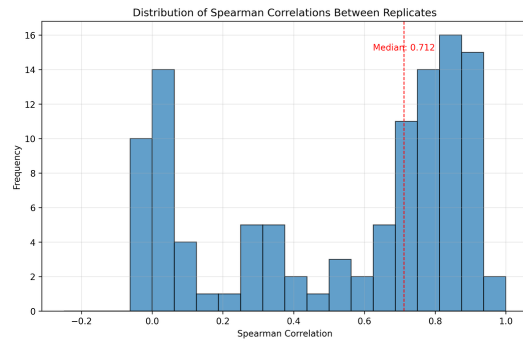


Figure 2: Spearman Correlation of Intensities Across Replicates

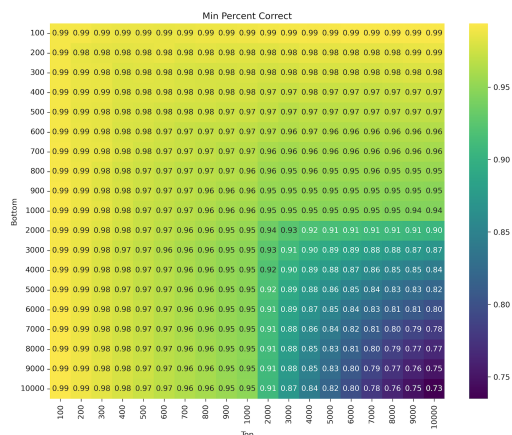


Figure 3: Average Accuracy Between Top-K and Bottom-K Buckets across Replicates

We collect a set of ~3,500 PBM experiments data from a publicly available database of PBMs, called CIS-BP (Catalog of Inferred Sequence Binding Preferences). Each experiment contains binding affinity for approximately ~40,000 DNA sequences, resulting in over 140 million unique binding interactions.

### 3.3.2 Initial Data Analysis

Before training, we performed a deep dive into the PBM data to curate an accurate and reliable dataset. In total, we had 4698 experiment files, each containing 1 protein binding domain and ~40,000 DNA sequences, spanning a total of 2633 protein binding domains. Within the dataset, we had 111 experiments that were functionally paired replicates, sharing the same protein binding domain and a significant overlap in tested DNA sequences. As these experiments offered a natural way to measure the noise within PBM experiments, we investigated the correlation between intensities of the same protein/dna pairs between replicates. Notably, we observed that intensities were not frequently preserved across replicates. The average spearman correlation across replaces was 0.712, and a non-negligible number of datasets had near zero or even negative correlation (Figure 2).

This lackluster correlation between replicates led us to question the accuracy of an individual intensity measurement. Thus, we investigated if broader trends in intensities would hold where individual readings do not. For each set of replicate experiments, we created buckets of the top-K and bottom-K DNA sequences in one replicate. We then compared the percentage of top-K DNA sequences that had intensities greater than the bottom-K sequences in the other replicate. We found that the top 1000 sequences very consistently had higher intensities than the bottom 1000 sequences, but this consistency degraded with larger buckets (Figure 3)

We also examined the distribution of intensities within an experiment. Notably, we see a number of experiments with an unusually large dynamic range of intensities. These aberrant experiments were



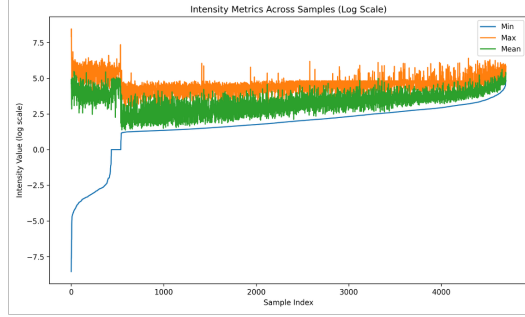


Figure 4: Intensity Ranges within Experiments

also associated with large negative intensity measurements, which are difficult to understand in the context of PBM experiments (Figure 4)

### 3.4 Experimental Setup / Training Scheme

For our final dataset, we performed the following filters SFT: to get a moderately sized dataset of high quality examples, we take the top 10 sequences for each PBM experiment. We do not filter on raw intensity values, as we are only taking the top sequences from each experiment and thus automatically ignore negative intensity values DPO: first, we filter out all experiments with negative intensities, removing untrustworthy data and datasets with abnormally high dynamic range. Then, for each experiment, we create buckets of the top 1000 and bottom 1000 sequences and generate preference pairs between one sample in the top bucket and one sample in the bottom bucket. Preference weights are generated by normalizing the relative intensities to sum to 1.

Our model is trained in two phases: supervised finetuning (SFT) and weighted direct preference optimization (DPO). When we create the model, the ESM LoRA layers, the DNA encoder transformer layers, and the cross-attention models are randomly initialized. Thus, the reference likelihoods from the base model are not meaningful, despite using pre-trained encoders. Thus, we use SFT to warmstart the model, tuning the new weights to adapt the foundation model embeddings for the task of protein-conditioned DNA generation.

For SFT, we finetune both the DNA encoder transformer layers and ESM LoRA layers. We combine two loss terms: an autoregressive causal language modeling loss to generate the top 10 DNA sequences per experiment from our dataset and a masked language modeling recreation loss on the protein sequence to ensure that our protein encoder does not lose its understanding of meaningful proteins. We equally weight the causal DNA loss and the masked protein loss and train for 84 epochs.

In order to perform DPO, we need reference likelihoods from our SFT model. To reduce memory bandwidth and avoid maintaining a reference model to generate likelihoods on-the-fly, we created a pipeline to precompute reference likelihoods across the entire DPO dataset using our SFT model. This was run across 700+ experiment files with 2000 sequences per file, with a runtime of 2.5 hours.

For DPO, we use the weighted DPO loss as described above. We continue to include the protein masked reconstruction loss to maintain the meaningfulness of the protein encoder. We use a beta value of 0.5 to give the model some slack to deviate from the base SFT model as necessary. For evaluation, we define the “validation top 1 accuracy” as the fraction of time that the model outputs a higher likelihood for the positive DNA sequence vs. the negative DNA sequence. Since our dataset of pairs is so large, we never finish an epoch. Thus, we periodically run evaluation on 200 random validation batches every 1000 training steps.

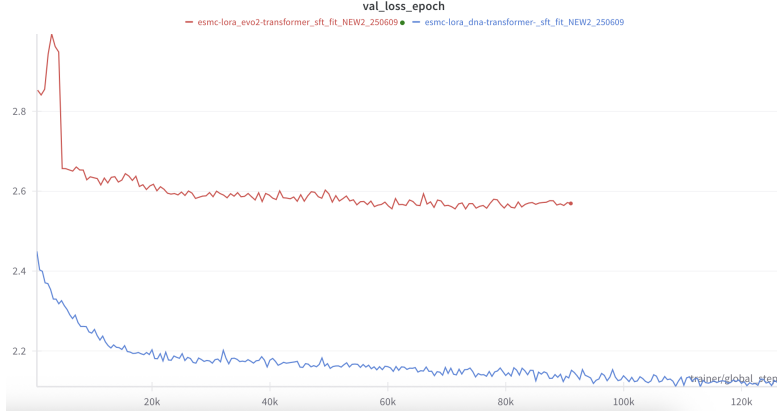


Figure 5: Loss Curves for Evo2 (red) and Custom Transformer (blue)

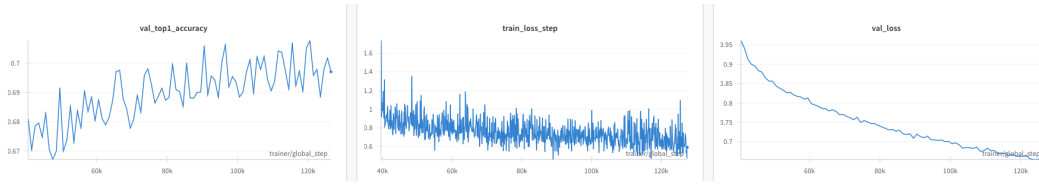


Figure 6: Loss Curves for DPO

## 4 Results

### 4.1 Quantitative Evaluation: Training

#### 4.1.1 SFT

Performing supervised finetuning of the base model proved to be a nontrivial task. We were able to low training and validation loss after fitting, but we observed that we had an exploding gradient norm.

We tried a number of mitigation strategies to try and clamp down on the exploding gradient norm, including implementing gradient clipping, lowering the learning rate, and training in bfloat16, but no method proved successful. Upon further investigation, we noticed that the output embedding values from Evo2 had a large dynamic range, spanning from  $1e-4$  to  $1e3$  in absolute magnitude. To mitigate this, we experimented with a dataset-wide normalization of the Evo2 dimensions before passing them through our transformer layers, but this also did not fix the gradient norm issue. Lastly, we noticed that if we removed the Evo2 trunk and simply tuned our transformer layers, we did not have any exploding gradient issues, leading us to believe that Evo2 was leading to instability.

In parallel, we decided to train our own DNA transformer model from scratch on the SFT data instead of using Evo2. This model was much more stable to train (no exploding gradient norm) and was able to achieve lower validation loss (Figure 5)

#### 4.1.2 DPO

After running DPO on our model, we observe the following training results. Both the training and validation loss curves decrease, showing that we are able to successfully learn on the DPO objective with our training pipeline. Additionally, we can see that the surrogate DPO loss is able to lead to real accuracy improvements beyond the plain SFT module as well (Figure 6). It is interesting that our accuracy curves are relatively noisy compared to the validation loss. One possibility is that we validate on a small subset of the data (200 random batches) for efficiency. Computing the accuracy over a fixed, larger validation set could offer a more stable view of performance improvement.

| Organism | Protein name | Protein sequence      | DNA start | DNA generation                   | Known binding | Match |
|----------|--------------|-----------------------|-----------|----------------------------------|---------------|-------|
| Human    | Nanog        | QKMRTVFSQAQLCALDKRFQ  | A         | ACTTAATTACGATACGATAATTAGTTGATT   | GCAATCA       | O     |
|          |              |                       | C         | CTAATCGATAATTGGTCGATATTTTGATCCC  |               | X     |
|          |              |                       | G         | GCTTAATTACGATACGATATTAGGTGATT    |               | O     |
|          |              |                       | T         | TTTACGATAATTGGTATTACGTAATTACGA   |               | X     |
| Human    | NKX2-5       | RKPRVLFSQAQVYELERRFK  | A         | ACTCTTAAGTGTITTTTCATGAGTTTTTAAT  | CACTCAA       | O     |
|          |              |                       | C         | CTCATTAAAGTGTITTTTCATGTTTTTTAATA |               | X     |
|          |              |                       | G         | GCTTAAGTGTCTTTTCATGTTTTTTTATA    |               | X     |
|          |              |                       | T         | TTTCAATGAGTGTITTCACCTAAGTTAATTAC |               | O     |
| Human    | Hoxb6        | RRGRQTYTRYQTLELEKEFH  | A         | ACTTAATTGCGTTTTTTAGTTTCATGGCTGA  | TAATTAC       | X     |
|          |              |                       | C         | CTGCTTAATTACTTTTTTTTTTTTATTTA    |               | O     |
|          |              |                       | G         | GGGGGGGGGGGGGGGGGGGGGGGGGGGGGG   |               | X     |
|          |              |                       | T         | TGCTTAATTACTTTTTTTTATTTTATTTA    |               | O     |
| Mouse    | Sox17        | IRRPMMNAFMVWAKDERKRLA | A         | ATTGTTTCATTGATCTTTCGGCAGGCTCTAA  | ATTGTC        | X     |
|          |              |                       | C         | CATTGTTTCATTGCTCATTGTGGGCACTAA   |               | O     |
|          |              |                       | G         | GATTGTTTCATTGCTCATTGTGGGCACTAAA  |               | O     |
|          |              |                       | T         | TTTTGTTTCATTGCTTTGGGTTGGGCTAAAC  |               | O     |
| Yeast    | Pho2         | RPKRTRAKGEALDVLKRKFEL | A         | AACTAATTAATTAATATATTAATTAGCTT    | ATAATA        | O     |
|          |              |                       | C         | CCCCCCCCCAGGGTAAAAACAAATATATA    |               | X     |
|          |              |                       | G         | GTTTAATTAATTAATTAATTAGCGGTTTTTA  |               | O     |
|          |              |                       | T         | TTTTAATTAATTAATTAATTAGCGTGACTTA  |               | O     |

60%

Figure 7: *De novo* DNA sequences from *ab initio* DNA transformer (SFT), conditioned on protein

## 4.2 Qualitative Analysis: Inference

Given limited time for a course project, we performed a preliminary evaluation using the SFT model only. However, given the positive training results during DPO, we expect similar, if not better performance from the DPO models. (For performance on the DPO models, be on the look out for a subsequent preprint soon!)

We used the *ab initio* DNA transformer model after SFT phase, to autoregressively generate *de novo* DNA sequences of length 30 base pairs, conditioned on 5 select, well-studied transcription factors across different protein families (Nanog, NKX, Hox, Sox, Pho) and different organisms (human, mouse, and yeast). For each protein, we generated 4 sequences, each starting with one of four nucleotides (A, C, G, T), given that Evo 2 does not use <BOS> tokens (and thus our DNA transformer also did not use this). We then matched the generated DNA sequences with their known DNA binding sequences from public databases (JASPAR, HOCOMOCO). Out of the 20 DNA sequences generated, we confirmed that 60% of the sequences recovered their known binding sequence, independent of protein family and organism (Figure 7). Additionally, being well-studied proteins, while the conditioning protein sequences were partially observed during training, the generated DNA sequences did not match with any sequences in the training data, confirming their *de novo* generation.

While this is a small, preliminary evaluation, we plan to scale up to more generated sequences across a broader coverage of unseen proteins for the DPO models, and expect to see better performance.

## 5 Discussion

Thus far, we have shown a promising first step towards generating DNA sequences capable of binding to a query protein domain. However, our method has a few limitations. First, we are not close to scaling our model training to the data that is available to us. Our training was limited to one GPU, and a full day of training covers less than 1% of the preference data in our dataset. Multi-GPU and distributed training would allow us to better leverage our data and improve performance. Still, our work shows that this method holds promise, and deep learning techniques are able to recover known DNA binding motifs from noisy and unannotated data.

We also encountered a number of difficulties while working on this project. First and foremost, we found that finetuning Evo 2 led to large training instabilities that proved difficult to overcome. Further work is needed to properly finetune Evo 2 and evaluate its performance. Second, our work highlighted flaws in the current state of open source biological foundation models. While ESM-C was easy to use and integrate into our codebase, Evo 2 required us to overcome many hurdles. Installation required many low-level dependencies that were difficult to install, and even afterwards, the inference-only model of Evo 2 was the only one that was accessible to us. For rapid scientific progression in the future, a more ergonomic implementation is needed.

## 6 Conclusion

In this work, we show that deep learning is an effective and promising method to better understand protein-DNA binding interactions. By leveraging protein binding microarray data, we demonstrate that biological foundation models combined with weighted DPO are able to learn biologically meaningful motifs and prioritize known DNA binding sequences. As this method continues to scale and utilize the full corpus of PBM data that exists, we expect the performance of this method to continue improving. In time, this method could allow biologists to design custom DNA sequences to recruit targeted transcription factors to specific genes in specific celltypes, opening the door for new gene therapy techniques that have higher efficacy, higher safety, and lower off-target side effects.

## 7 Team Contributions

The work for this project was evenly distributed. To highlight some individual contributions:

- **Group Member 1:** Chang M. Yun: model implementation, exploding gradient debugging, writing
- **Group Member 2:** Ryan Zhao: training pipeline implementation, writing, data module integration, likelihood preprocessing
- **Group Member 3:** Salil Deshpande: data deep dive, data wrangling and loading, SFT training implementation

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